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Autonomous biomedical research with an artificial intelligence agent

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Biomedical research is increasingly constrained by repetitive, fragmented workflows that slow discovery. We introduce Biomni, a general-purpose biomedical artificial intelligence agent that autonomously executes diverse research tasks. To map the biomedical action space, Biomni's action-discovery agent mines tools, databases, and protocols from thousands of publications across 25 domains, building a unified agentic environment. Its general-purpose architecture integrates large language model reasoning with retrieval-augmented planning and code-based execution, dynamically composing workflows without predefined templates. Systematic benchmarking shows strong generalization across heterogeneous tasks—causal gene prioritization, drug repurposing, rare-disease diagnosis, microbiome analysis, and molecular cloning—without task-specific tuning. Real-world case studies demonstrate Biomni interpreting multi-modal datasets, optimizing protein stability, orchestrating wet-lab instruments, and generating experimentally testable protocols. Biomni envisions artificial intelligence augmenting human scientists and accelerating discovery.

Biomedical research is a key pillar of modern science and medicine, driving discoveries in disease mechanisms, diagnostics, and therapeutics (*1*–*4*). Yet, with the growth in large-scale experiments, data, tools, and literature, progress is increasingly slowed by fragmented, complex workflows that make it difficult to efficiently integrate specialized tools, exhaustive literature reviews, intricate experimental design, and rigorous statistical modeling (*5*, *6*). A vast volume of valuable biomedical data sits underutilized (*7*), many sophisticated analyses are not conducted, and many connections for past knowledge and literature are not made, not for lack of value, but because the demand for expert researchers far exceeds the supply. This mismatch between data abundance and limited human bandwidth highlights an urgent need for approaches that can effectively scale expertise, streamline workflows, and unlock the full potential of biomedical research.

Artificial intelligence (AI) agents have dramatically reshaped fields such as software engineering (*8*), law (*9*), material science (*10*) and healthcare (*11*) by automating repetitive tasks, enhancing productivity, and enabling capabilities that were previously difficult to achieve. The development of AI agents also offers opportunities to reshape aspects of biomedical research (*12*). We envision such agents tackling diverse biomedical research tasks across subfields, thus augmenting human biologists' specialized expertise. Capable of efficiently managing thousands of concurrent tasks, AI agents could enhance human productivity and accelerate the pace of biomedical discovery.

Previous and concurrent biomedical AI agents are largely specialist systems built for narrow tasks (*13*–*20*), limiting their ability to generalize across biomedical domains. Virtual Lab orchestrates multi-agent LLM workflows for SARS-CoV-2 nanobody design (*14*); BioDiscoveryAgent runs closed-loop gene-perturbation selection within predefined screens (*15*). CellAgent autonomously configures and executes scRNA-seq analysis pipelines (*17*), and scBaseCount uses agents to curate and continually expand a single-cell data repository (*21*) — both powerful but confined to single-cell omics. More recent efforts extend agents to additional modalities and tasks (*22*): SpatialAgent applies multi-agent reasoning to spatial transcriptomics for spatially aware inference and hypothesis

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exploration (*23*), while parallel advances in self-evolving, reinforcement-learning-driven, and automated hypothesis-generation architectures show promise (*24*), yet each remains restricted to a task-specific action space (*25*).

Enabling an AI agent to handle a broad range of biomedical tasks introduces substantial technical challenges – most notably, the need to tightly couple advanced reasoning (*26*) with the ability to execute highly specialized biomedical actions (*27*). Although LLM-based reasoning has seen major advances (*28*), such LLMs need access to an environment that explicitly defines the biomedical action space, which is inherently diverse, domain-specific, and complex. Moreover, a truly capable system requires an agentic architecture that can natively interact with this biomedical environment to select and execute diverse tasks without relying on pre-defined workflows.

Here we present Biomni, a general-purpose biomedical AI agent purpose-built to assist and partially automate biomedical research across a wide range of subfields. Biomni formulates hypotheses, performs complex bioinformatics analyses, and designs rigorous experimental protocols. To enable this capability, we first constructed a unified and comprehensive biomedical action space by systematically analyzing 2,500 biomedical research papers spanning 25 distinct subfields, curated from literature repositories. From this foundation, we developed an LLM-powered action discovery agent capable of reading papers and extracting key tasks, tools, and databases essential to driving biomedical discoveries. These elements are then selected and implemented into Biomni-E1, the foundational environment that defines the biomedical action space for agentic interaction. Biomni-E1 includes 150 specialized biomedical tools, 105 software packages, and 59 databases. We then designed Biomni-A1, a general-purpose agent architecture capable of flexibly executing a broad spectrum of biomedical tasks by using tools and datasets provided by Biomni E1. Given a user query, the agent first uses a retrieval system to identify the most relevant tools, databases, and software needed. It then applies LLM-based reasoning and domain expertise to generate a detailed, step-by-step plan. Each step is expressed through executable code, enabling precise and flexible compositions of biomedical actions – an essential feature given the domain's reliance on highly specialized tools and data resources. This integrated system allows Biomni to generate solutions for challenging, large-scale biomedical problems with efficiency, but also to generalize to tasks across previously unseen areas of biomedical research.

Rigorous benchmarking demonstrates Biomni's strong performance across established biomedical quality and assurance benchmarks, and robust generalization performance in challenging, realistic scenarios never encountered during development. We highlight Biomni's practical capabilities through five case studies: (1) analyzing wearable sensor data; (2) performing comprehensive bioinformatics analyses on massive raw datasets such as single-cell RNA-seq and ATAC-seq data; (3) designing laboratory protocols to assist wet-lab researchers; (4) optimizing a protein sequence for better thermostability; (5) orchestrating robotics wet-lab instruments. With Biomni, we introduce a scalable, general-purpose biomedical AI agent, pointing toward a future in which AI agents work alongside human researchers to accelerate biomedical discovery from basic research to translation.

Overview of Biomni

Curating a unified biomedical action space is challenging due to its inherent complexity and vastness. We systematically address this problem by employing an AI-driven approach (Fig. 1A). We leveraged the 25 subject categories defined by bioRxiv, selecting the 100 most recent publications per category. An action discovery LLM agent processed each paper sequentially, extracting essential tasks, tools, databases, and software necessary to replicate or generate the described research. This comprehensive set of resources constitutes the essential actions required to perform a large set of biological research tasks.

We then curated Biomni-E1, an environment for a biomedical AI agent to perform a wide range of actions (Fig. 1B). Identified tools were rigorously verified by human experts, along with corresponding test cases. These tools (tables S1 to S18) were specifically chosen for their non-trivial nature, encompassing complex code, domain-specific know-how, or specialized AI models. Recognizing the inherent flexibility required by biological software, which cannot always be simplified into static functions, we constructed an execution environment pre-installed with 105 widely used biological software packages (tables S24 to S31), supporting Python, R, and CLI. For database integration, we categorized resources into two distinct groups. The first group consists of massive relational databases accessible via web APIs (e.g., PDB, OpenTarget, ClinVar) (tables S19 and S20). Rather than creating numerous individual retrieval tools, we implemented a unified function per database. Each function accepts natural language queries and internally employs an LLM to parse database schemas and generate executable queries dynamically. Databases without web interfaces were downloaded into a data lake and preprocessed locally into structured flat files for seamless integration with the agent, for a total of 59 databases in Biomni-E1 (tables S21 to S23).

To build a general-purpose agent capable of tackling diverse biomedical tasks, we require a specialized agentic architecture – one that avoids hardcoding workflows for each individual task. This led to the development of Biomni-A1, which incorporates several core innovations critical for operating across the biomedical research landscape. First, we

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introduce an LLM-based resource selection mechanism designed to navigate the complexity and specialization of biomedical environment, dynamically retrieving a tailored subset of resources based on the user's goal (fig. S1 for ablation experiment on the retrieval step). Second, recognizing that biomedical tasks often require rich procedural logic, Biomni-A1 uses code as a universal action interface – allowing it to compose and execute complex workflows involving loops, parallelization, and conditional logic. This approach also enables the agent to interleave calls to software, tools, databases, and raw data operations that do not conform to predefined function signatures, supporting flexible and dynamic integration of heterogeneous resources. Third, the agent adopts an adaptive planning strategy: it formulates an initial plan grounded in biomedical knowledge and iteratively refines it throughout execution, enabling responsive, context-aware behavior. Together, these innovations enable Biomni-A1 to generalize to previously unseen tasks and domains, dynamically composing actions, and interfacing with software, data, and tools (Fig. 1C).

Biomni excels on general biomedical research benchmarks

To establish baseline competence across a broad range of biomedical research problems, we first evaluate Biomni on standardized benchmarks that span common biological tasks (Fig. 2). We first evaluate performance on Biomni-Eval1, which consists of 443 queries spanning 10 representative biomedical research tasks (Fig. 2A): CRISPR delivery, causal gene detection (gene-centric, ontology-based, and pathway-based), variant prioritization, database query, DNA sequence query, patient gene detection, rare disease diagnosis, and perturbation screen design. Across these tasks, Biomni achieves an average accuracy of 57%, outperforming a base LLM (Claude Sonnet 4.5) (30%), a therapeutics-specific agent called TxAgent (25%), a general coding agent called Claude Code (43%), and a bioinformatics software-based agent using ReAct equipped with the Biomni-E1 (44%).

We further evaluate general biomedical reasoning using Humanity's Last Exam (HLE-Bio) (Fig. 2B) to test robustness across unfamiliar domains without task-specific tuning. Across multiple frontier LLM backbones, the Biomni agent scaffold yields consistent improvements over LLM-only baselines, with absolute accuracy gains of 6-12 percentage points, indicating that the observed improvements are not driven by a particular underlying model but by the Biomni A1 agent scaffold and E1 environment. Additional trajectory and error analysis are available in figs. S6 to S17.

Biomni performs biomedical research tasks at expert level while shortening the time

Having established strong performance on benchmarks, we next assessed whether these gains translate to realistic end-to-end biomedical research workflows by comparing with human experts. We evaluated Biomni on three representative tasks, single-cell RNA-seq annotation, rare disease diagnosis, and GWAS causal gene detection, with direct comparison to human expert performance (Fig. 2, C to E; see materials and methods and Data, code, and materials availability statement for prompts and responses). Experts were recruited based on having at least five years of experience in the relevant task domains and included postdoctoral researchers and faculty members from academic institutions. The prompt was developed by the authors. In single-cell annotation (Fig. 2C), Biomni achieves 45.8% accuracy, in the range of the strong expert baseline (40.5% and 50.9%), while reducing average analysis time from approximately 230 min to 75 min per dataset. For rare disease diagnosis (Fig. 2D), Biomni attains 60% accuracy across cases, matching 5 expert performance (60%-70%) while completing analyses in roughly 3 min, compared to over 110 min for human experts. In GWAS causal gene detection (Fig. 2E), Biomni reaches 80% accuracy across queries, again comparable to expert performance, while reducing execution time from approximately 90 min to 4 min per locus. Across tasks, Biomni consistently matches expert accuracy while requiring substantially less time.

Reinforcement learning enables scalable improvement on specialized biomedical tasks

Despite strong generalization, performance on several specialized tasks remains below expert level, and qualitative analysis shows that models do not consistently exploit the full capabilities of the Biomni-E1 environment (see fig. S8), motivating a reinforcement learning approach to scalably improve task-specific performance. Figure 2, F to H, introduces Biomni-R0, a reinforcement learning (RL)-trained open-source model that optimizes end-to-end task success through direct interaction with the Biomni-E1 environment. Using expert-annotated rewards, RL training substantially improves average task performance: Biomni-R0-8B increases from 0.32 to 0.59, and Biomni-R0-32B increases from 0.35 to 0.67 (Fig. 2G). Biomni-R0-8B exceeds the performance of larger closed-source models such as Claude 4 Sonnet (0.56), and scaling to 32B yields an additional 10% absolute gain. Task-level results in Fig. 2H show consistent improvements across CRISPR screen design, causal gene detection, variant prioritization, database querying, and rare disease diagnosis, indicating that reinforcement learning provides a practical mechanism for systematically hill-climbing performance on specialized biomedical agent tasks.

Biomni performs population-scale autonomic and chronobiological analysis from raw wearable data

To evaluate Biomni's performance in real-world biomedical

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workflows, we invited scientists to apply it directly to their own research questions. As biomedical research spans across a wide range of tasks, we select 5 use cases for illustrative purpose, with additional 4 use cases available in figs. S2 to S5.

We first applied Biomni to two large, previously published wearable datasets with gold-standard human analyses. In one benchmark, we used the COVID-19 physiological response cohort from Alavi *et al.* (*29*), consisting of raw minute-resolution Fitbit heart-rate and step data from 1,027 participants, comprising over 1.4 billion heart-rate measurements and 37 million step records, with a mean monitoring duration of 170 days per individual. The dataset is highly heterogeneous, containing long-term longitudinal recordings across individuals with diverse baseline physiology and activity patterns.

Starting exclusively from raw heart rate and step data, Biomni autonomously generated and executed a complete end-to-end analysis pipeline (see materials and methods, section G, and the Data, code, and materials availability statement). The agent extracted six validated COVID-19-associated physiological biomarkers, including elevated resting heart rate, reduced circadian amplitude, decreased heart-rate variability, elevated nocturnal heart rate, reduced daily steps, and blunted chronotropic response. It further reconstructed population-level 24-hour circadian structure, integrated biomarkers into a multi-dimensional risk score (0-6), and generated publication-quality visualizations. The resulting biomarkers, effect sizes, and correlations closely match those reported in the original study and subsequent literature, including a negative correlation between resting heart rate and circadian amplitude ($r = -0.34$) and a positive correlation between daily steps and heart-rate variability ($r = +0.54$). These results indicate that Biomni can independently reproduce expert-level autonomic and chronobiological analyses directly from raw wearable data at population scale.

Biomni automates complex multi-omics analysis to decipher transcriptional regulation of skeletal lineages

To test whether Biomni could generalize to complex omics workflows, we used it to analyze a recently published multi-omics dataset of the developing human skeleton (*30*). This dataset comprises 336,162 single-nucleus RNA-Seq (snRNA) and ATAC-seq (snATAC-Seq), paired with spatial transcriptomics data collected from human embryos between 5-11 weeks post-conception (Fig. 3E). While the original study emphasized developmental trajectories and disease mechanisms, we were interested in exploring gene regulatory mechanisms across emerging skeletal cell types – a technically demanding task typically requiring extensive bioinformatics support.

We asked Biomni to investigate transcriptional regulation across skeletal lineages using a detailed instruction (see materials and methods, section H). Biomni autonomously planned and executed a ten-stage analysis pipeline: (1) loading and exploring all datasets, (2) preparing RNA-seq data for analysis, (3) configuring pySCENIC to retrieve motifs, (4) running GRNBoost2 to infer gene regulatory networks, (5) pruning networks using cisTarget, (6) calculating regulon activity with AUCell, (7) extracting accessibility data from ATAC-seq, (8) filtering predicted targets using ATAC-seq accessibility, (9) analyzing activity patterns across cell types, developmental stages, and anatomical regions, and (10) summarizing findings and preparing a report. This pipeline output transcription factor-target gene links and filter regulons based on motif enrichment and chromatin accessibility correlations (Fig. 3F).

The full run, completed in around five hours, handled real-time execution issues (e.g., variable name mismatches) by subsampling and debugging locally. Throughout the run, Biomni maintained all intermediate outputs – code, figures, and logs – organized in a reproducible notebook structure for validation and inspection (see Data, code, and materials availability statement). The agent summarized all the analysis and generated a report describing the analysis and key findings (see materials and methods, section H).

In its final gene regulatory network (GRN) analysis (Fig. 3H), Biomni re-capitulated known regulatory relationships between key osteogenic transcription factors such as RUNX2 and HHIP, confirming how they are regulated by a shared set of anti-osteogenic transcription factors including TWIST1, LMX1B, and ALX4 (*30*). These findings align with author's report about the balanced regulation needed for proper bone formation and suture patency (*30*). Furthermore, Biomni nominated several putative regulators that have not been prominently highlighted in prior human skeletal GRN studies, including AUTS2, ZFHX3, and PBX1, which showed high inferred regulatory activity across multiple skeletal cell types. This signal is supported by complementary evidence from motif enrichment and TF footprinting, concordant AUCell regulon activity, ATAC co-accessibility of predicted targets, and cross-region and cross-stage reproducibility, with explicit decision criteria, negative controls, and per-TF evidence summarized in a new table (Fig. 3, F to H, and Discussion). PBX1 is a well-established skeletal regulator (*31*) and ZFHX3/AUTS2 have only limited or indirect skeletal reports (in mouse (*32*) or zebrafish (*33*)), their broad activity here suggests under-appreciated roles across diverse skeletal lineages. Biomni reported that these novel regulators were particularly active in osteoblasts, preosteoblasts, and various chondrocyte populations, suggesting they play important but previously unrecognized roles in the transcriptional control of skeletal cell fate determination during human skeletal development. Finally, Fig. 3, G and H, reveals how Biomni's visualizations effectively captured both temporal dynamics of

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regulator activity and cell-type-specific variations in key regulons like RUNX2. This example demonstrates how Biomni enables researchers to autonomously perform complex multi-omics analysis and rapidly generate testable hypotheses without specialized programming expertise.

Biomni designs wet-lab validated experimental protocol for cloning

To evaluate Biomni's ability to support real-world experimental design, we focused on a core task in molecular biology: cloning. This process is central to countless workflows in research and biotechnology and requires complex reasoning, from designing high-fidelity primers to choosing the right assembly method and validating constructs. Whereas general-purpose LLMs have struggled to perform such tasks due to limited domain knowledge and tool access (*34*), Biomni integrates LLM reasoning with dynamic tool execution, enabling expert-level performance in molecular biology tasks.

To rigorously evaluate this task, we first collaborated with an expert group of gene-editing researchers to design an open-ended cloning benchmark and expert user study (Fig. 4A). Our benchmark consisted of 10 realistic, representative cloning tasks covering Golden Gate, Gibson, Gateway, and restriction cloning – each with options including single-fragment vs. pooled assembly. The benchmark also included essential validation steps, such as designing Sanger sequencing primers and analyzing restriction digests. We posed these tasks to four entities: an LLM, Biomni, a human trainee (Stanford Biology graduate with previous experience in cloning, S.Z.), and a senior human expert (Stanford Genetics postdoctoral researcher with 5+ years of cloning experience, D.Y.). Each was asked to generate a complete, end-to-end protocol along with the final cloned plasmid map. Blinded expert reviewers assessed the outputs. Biomni produced protocols and designs that matched the human expert in accuracy and completeness as determined by manual examination of the outputs by an independent expert. Biomni often provided comparable levels of detail and anticipated the same edge cases. In contrast, the human trainee's submissions were frequently incomplete or suboptimal, reflecting the experience gap typical in early-stage researchers. Remarkably, Biomni completed all tasks autonomously in a fraction of the time taken by the expert.

To further validate Biomni in a real-world setting, we assigned it a practical cloning task: cloning a guide RNA targeting the human B2M gene into the lentiCRISPR v2 Blast construct (Fig. 4B). Biomni successfully executed the task through a comprehensive workflow (Fig. 4C). First, it analyzed the plasmid structure using annotation and pattern search tools to identify key features necessary for cloning. It then designed three Cas9 sgRNAs targeting B2M using specialized knockout sgRNA design tools. For the cloning process, Biomni generated forward and reverse oligos with BsmBI overhangs to enable directional insertion of the sgRNA sequence. It produced detailed protocols (Fig. 4D) for oligo annealing, double-stranded DNA formation, and Golden Gate cloning into the target vector. Biomni also provided complete bacterial transformation instructions, including heat-shock steps and antibiotic selection. For quality control, it designed a U6 promoter sequencing primer to verify sgRNA insertion and simulated the Golden Gate assembly to produce the final plasmid map (see materials and methods, section I).

We followed Biomni's protocol exactly to perform the wet-lab experiment (Fig. 4E). Colonies appeared on the plate the next day; two were cultured, miniprepped, and sequenced using the Biomni-designed primers – both showing perfect alignment. This case illustrates how scientists can rely on Biomni to autonomously design complex molecular biology experiments with accuracy comparable to human experts, but in a fraction of the time.

Biomni orchestrates AI models for iterative molecular design

To demonstrate Biomni's capacity for autonomous molecular design, we integrated state-of-the-art protein modeling tools-including AlphaFold-2 (*2*), ThermoMPNN (*35*), and generative design models - enabling researchers to leverage these powerful but infrastructure-intensive methods through natural language alone. In this case study, we prompted Biomni with a protein sequence and the objective of improving its thermostability. Without any coding, GPU configuration, or specialized computational expertise, Biomni autonomously constructed and executed an optimization pipeline (Fig. 5A): it predicted the initial 3D structure using AlphaFold-2, assessed baseline thermostability via ThermoMPNN, searched literature for stabilizing mutations in homologous proteins, proposed candidate mutations informed by both structural analysis and domain knowledge, and iteratively evaluated each variant across three optimization cycles. The agent ultimately identified three mutations, Q83I, C66F, and C110F, yielding a cumulative predicted thermostability improvement of −4.108 kcal/mol while preserving 98% sequence identity (Fig. 5B). Each mutation aligned with established protein engineering principles: the Q83I substitution enhanced hydrophobic core packing, while the two cysteine-to-phenylalanine changes introduced aromatic stabilization through π–π stacking interactions. By serving as an intuitive interface to complex AI models that would otherwise require extensive infrastructure and expertise, Biomni democratizes access to sophisticated computational protein engineering.

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Biomni bridges computational analysis and wet-lab execution through automated protocol generation

A persistent bottleneck in biological research is the disconnect between computational design and physical experimentation: insights from in silico analysis must be painstakingly translated into laboratory protocols, often requiring days of manual coordination. To address this gap, we integrated Biomni with PyLabRobot (*36*), the open-source Python framework for laboratory automation, enabling researchers to generate and execute production-ready liquid handling protocols directly from natural language. We validated this capability on Hamilton STAR systems in production environments. Given a simple prompt describing a liquid transfer, Biomni generates complete, executable code with proper deck configuration, error handling, and resource cleanup (Fig. 5C). For complex workflows, we tested Biomni on a cell viability assay measuring dose-response of 8 compounds across 12-point serial dilutions in triplicate. From a hardware specification, Biomni produced a comprehensive protocol with intelligent tip selection, mathematically correct serial dilutions, and volume validation (see materials and methods, section N). This integration demonstrates how Biomni can serve as an interface between experimental intent and executable automation code, an important step toward closing the loop between dry-lab analysis and wet-lab execution.

Discussion

Biomni is a powerful platform biomedical research, demonstrating robust generalization across diverse subfields and laying the groundwork for AI agents as integral collaborators in scientific discovery. Its zero-shot performance across complex tasks – including those in genetics, genomics, microbiology, immunology, pharmacology, and clinical medicine – underscores its potential to boost research productivity and accelerate discovery.

By automating complex, labor-intensive workflows, which normally require both expert knowledge and coding skills, Biomni enables researchers to redirect their efforts toward creative hypothesis generation, experimental innovation, and cross-disciplinary collaboration. In the context of target and drug discovery for biopharma, Biomni could be potentially applied to design workflows to prioritize targets, design perturbation screens, or repurpose drugs. In clinical settings, Biomni can support gene prioritization and rare disease diagnosis. For consumer health, Biomni supports integration of wearable data and multi-omics analyses for health monitoring and intervention.

Nonetheless, several limitations remain. While Biomni's unified environment spans a wide range of biomedical tools and databases, the evaluated tasks represent only a subset of the field, and key domains remain unexplored. In addition, in the action discovery agent, our decision to prioritize the most recent literature makes the agent appear timely, but risks overlooking foundational concepts and techniques that have faded from current discourse despite their enduring relevance. Future versions should consider a larger range of publications when defining the environment. In addition, complex multi-step analyses, such as the scRNA-scATAC multi-omics workflow, often benefit from more structured prompts that explicitly define intermediate analytical steps. Although high-level prompts are sufficient for many tasks, workflows requiring substantial domain knowledge and implicit analytical conventions achieve greater robustness and reproducibility when key steps are specified, suggesting the limitation on generalizability of the current AI agent. Moreover, although Biomni approaches human-level performance in tasks like database querying, sequence analysis, and molecular cloning, it still struggles in areas requiring nuanced clinical judgment, experimental reasoning, or deep biological thinking and synthesis. No system yet captures the full scope of human biomedical expertise. As reflected in our benchmarks, Biomni has not achieved expert-level performance across all task categories. Performance remains uneven across the diverse landscape of biomedical tasks, and Biomni does not yet perform strongly across the board. Because the space of possible biomedical applications is vast and manuscript space is limited, we present five representative use cases spanning diverse biological domains to illustrate the range of Biomni's capabilities rather than exhaustively covering all potential use cases. We expect continued improvements as foundation models evolve and the agentic environment expands, as well as thanks to human experts and trainees deploying Biomni to facilitate or augment their work.

These limitations open promising directions for future development. Training biomedical reasoning agents with reinforcement learning could enable continuous self-improvement in planning and execution. Integrating multimodal data – text, images, and structured inputs – may further deepen reasoning capabilities. Equipping Biomni to autonomously discover and incorporate new tools and databases, as well as incorporate more historical methods (which may have high utility but can be easily forgotten by human users), would ensure adaptability and long-term relevance.

Finally, we recognize that increasingly capable AI systems in the life sciences raise important biosecurity considerations. Although Biomni is designed to accelerate beneficial biomedical research, agentic systems capable of literature synthesis, protocol generation, and automated analysis could also lower barriers to the misuse of biological knowledge. Responsible development therefore requires careful evaluation of both risks and benefits. In this work, we adopt several principles to mitigate potential concerns: focusing the system on widely used academic datasets and tools, emphasizing

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transparency through open-source release, and aligning our development practices with emerging community discussions and policy frameworks on AI-biosecurity risks. We believe that openness, rigorous evaluation, and engagement with the broader biosecurity and policy community are essential to ensure that scientific AI agents are developed in ways that maximize societal benefit while minimizing potential harms.

Looking ahead, Biomni and its successors could become foundational infrastructure in an AI-powered biomedical ecosystem, working seamlessly with human experts to unlock insights into health and disease. This hybrid partnership may reshape biomedical research – automating hypothesis generation, scaling discovery pipelines, and enabling medical innovation to advance faster than before.

Materials and methods are available in the supplementary materials.

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were used for manuscript editing (ChatGPT) and to assist with code generation (Claude Code). All AI-assisted manuscript text and code were reviewed and verified by the authors. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S17
Tables S1 to S33
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MDAR Reproducibility Checklist

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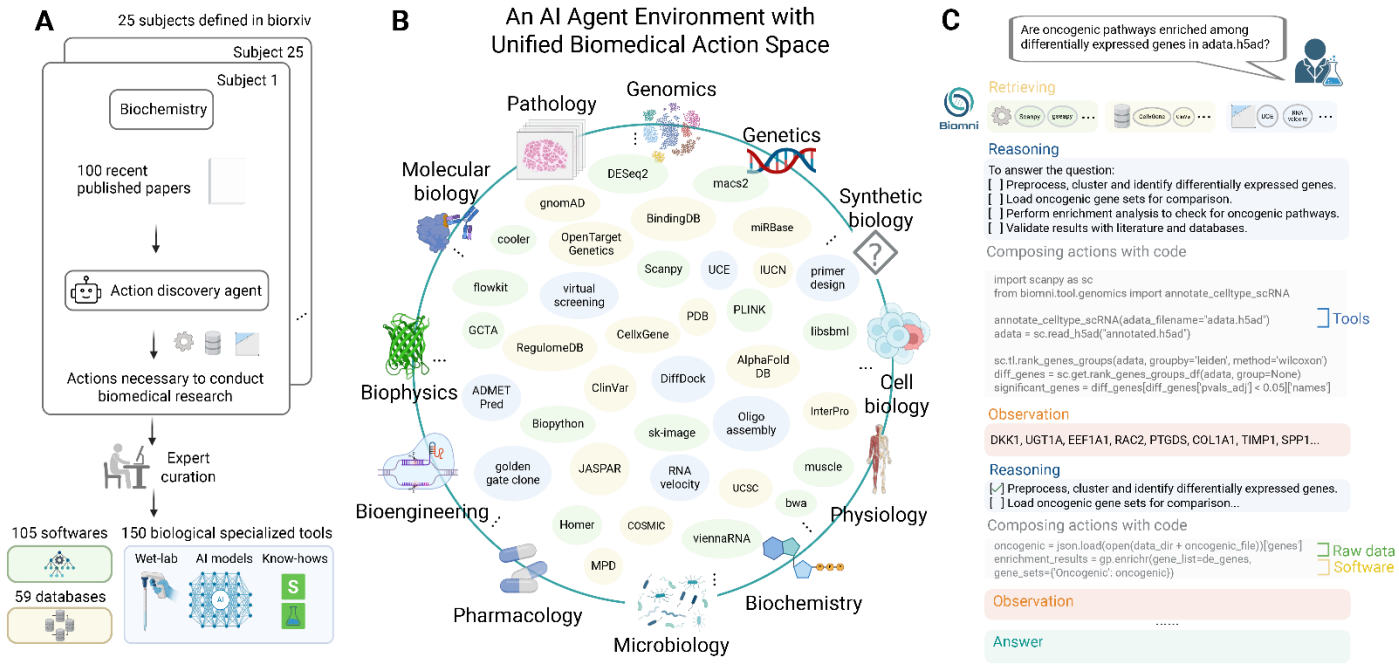


Fig. 1. Overview of the unified biomedical action space and agent environment in Biomni. (**A**) Workflow for systematically curating the unified biomedical action space. Actions necessary to conduct biomedical research were extracted from 2,500 recent bioRxiv publications across 25 biomedical subfields using an AI-driven discovery agent. Extracted actions were rigorously validated and curated by human experts, resulting in the integration of 105 biomedical software packages, 150 specialized biological tools (including wet-lab protocols, AI-driven predictive models, and domain-specific know-how), and 59 comprehensive biomedical databases. (**B**) Illustration of the unified biomedical action space spanning diverse biomedical subfields such as genetics, genomics, synthetic biology, cell biology, physiology, microbiology, pharmacology, bioengineering, biophysics, molecular biology, and pathology. Representative tools and databases integrated into Biomni's environment are shown, highlighting its general-purpose capabilities. (**C**) Example workflow demonstrating Biomni's reasoning and action composition process to autonomously answer a complex biological question. Biomni retrieves relevant tools based on the user's query, formulates a structured reasoning plan, and composes executable code to perform comprehensive bioinformatics analyses, iteratively refining its reasoning based on observations until converging on a final, precise answer.

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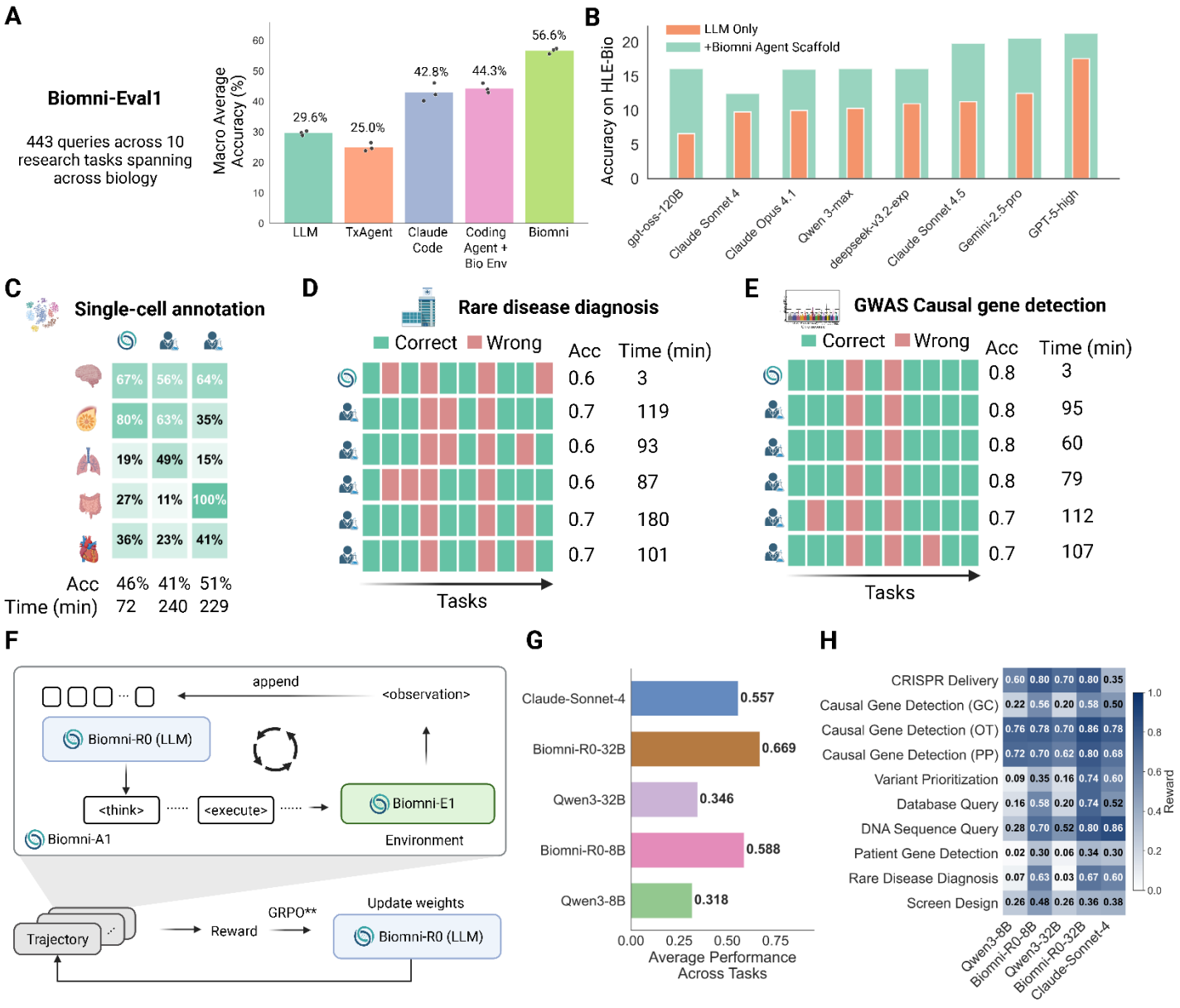
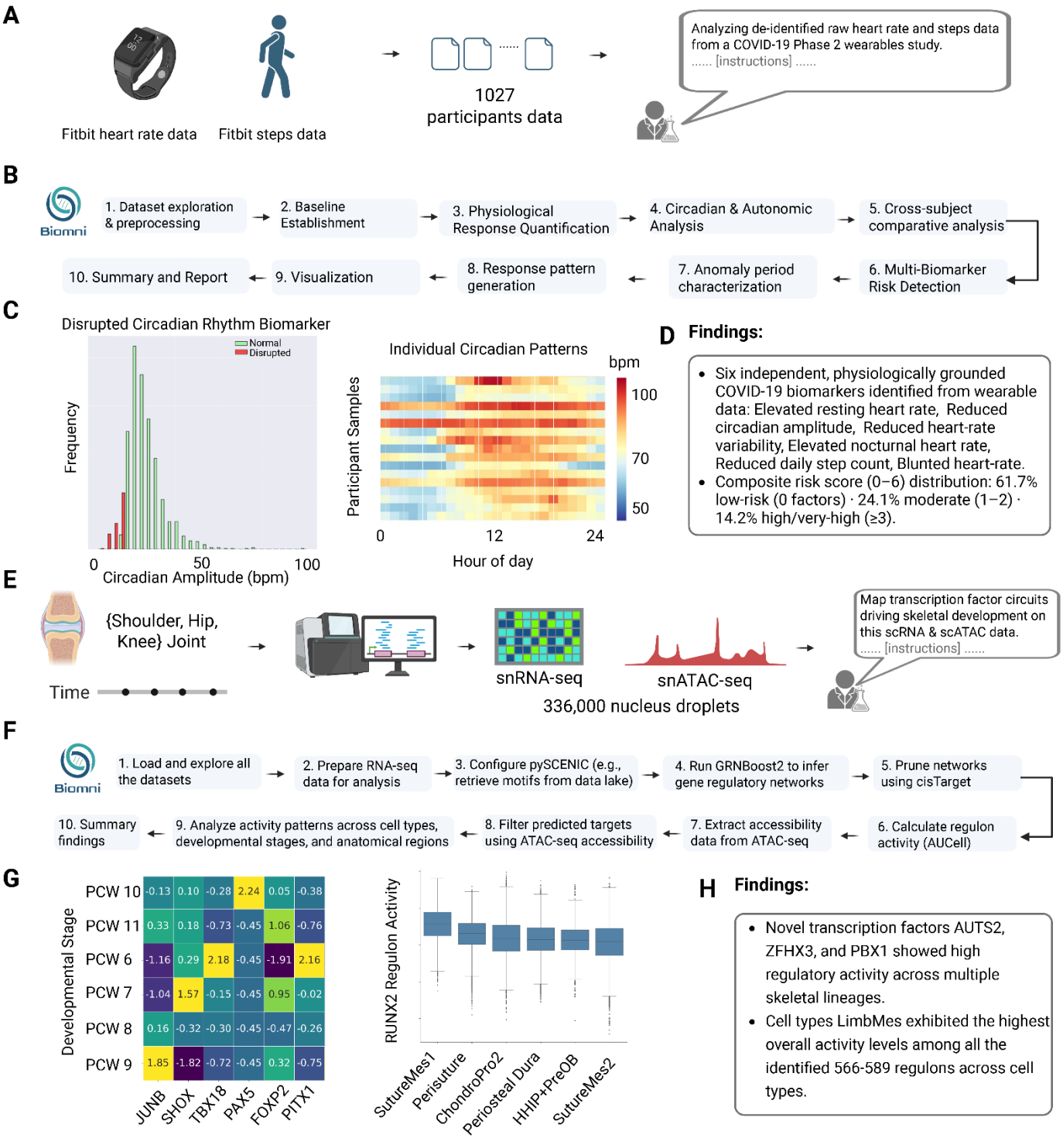


Fig. 2. Biomni performance across general benchmarks, expert-level tasks, and reinforcement learning. (**A**) Performance on Biomni-Eval1, comprising 443 queries across 10 biomedical research tasks, including CRISPR delivery, causal gene detection, variant prioritization, database and DNA sequence queries, patient gene detection, rare disease diagnosis, and screen design. The data points are 3 independent runs, and we report the average accuracy across 443 queries. (**B**) Accuracy on Humanity's Last Exam (HLE-Bio) across multiple frontier LLM backbones, comparing LLM-only baselines with the Biomni-A1 agent scaffold and Biomni-E1 agent environment. (**C**) Single-cell RNA-seq annotation accuracy and execution time compared with human experts, showing expert-comparable accuracy with reduced analysis time. (**D**) Rare disease diagnosis accuracy and execution time, with Biomni matching expert accuracy while substantially reducing time to completion. (**E**) GWAS causal gene detection accuracy and execution time, demonstrating expert-level performance with markedly faster execution. (**F**) Overview of the Biomni agent architecture and reinforcement learning training procedure. (**G**) Average task performance before and after reinforcement learning, showing substantial gains for Biomni-R0 models at 8B and 32B scales. Biomni-R0 was fine-tuned from Qwen3 open-weight model (*37*) and was first distilled from the frontier model Claude-Sonnet-4 (i.e., teacher model) and then performed reinforcement learning to further specialize to biomedical tasks (see materials and methods). It outperformed the teacher frontier model after tuning. (**H**) Task-level performance comparison across specialized biomedical tasks, illustrating consistent improvements from reinforcement learning across task categories.

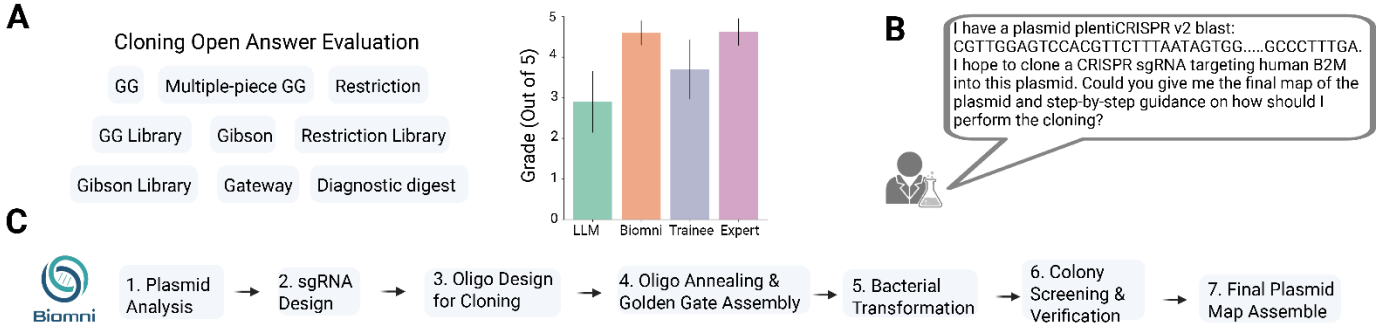
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Fig. 3. Biomni autonomously executes complex multi-modal biomedical analyses to generate hypotheses. (**A** to **D**) Overview of the COVID-19 Phase 2 Wearables Study and analysis workflow. (A) High-frequency heart-rate and step-count data were collected via Fitbit devices from 1,027 participants during the COVID-19 pandemic. Data were processed to establish individual physiological baselines and detect illness-related deviations. (B) 10-step analytical pipeline applied to the full cohort using Biomni software. (C) Key figures generated by Biomni, including population-level heart-rate and step-count distributions during COVID-19 positive periods (left) and individual circadian pattern heatmaps (right). (D) Biomni identified key physiological correlations confirm interconnected mechanisms. (**E** to **H**) Biomni autonomously analyzed single-cell multiomics data from approximately 336,000 nucleus droplets, combining single-nucleus RNA (snRNA-seq) and single-nucleus ATAC sequencing (snATAC-seq) across human embryonic joint development (shoulder, hip, knee). (F) A detailed workflow diagram showing Biomni's 10-step analysis pipeline for gene regulatory networks with multiomics. (G) Two key figures generated from Biomni: Left panel shows a heatmap of regulator activity by developmental stage, with color intensity indicating activity levels. Right panel displays a boxplot of RUNX2 regulon activity by cell type, showing variation in expression across different cell populations.

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D Biomni generated wet-lab cloning protocol

Step-by-Step Cloning Protocol

1. Design and Order Oligos
- **Forward oligo**: 5'-CACCGCGCGAGCACAGCTAAGGCCA-3'
- **Reverse oligo**: 5'-AAACTGGCCTTAGCTGTGCTCGCGC-3'

2. Anneal and Phosphorylate Oligos
1. Resuspend oligos to 100 µM in nuclease-free water
2. Prepare annealing reaction:
   - 1 µL Forward oligo (100 µM)
   - 1 µL Reverse oligo (100 µM)
   - 1 µL 10X T4 Ligation Buffer
   - 6.5 µL nuclease-free water
   - 0.5 µL T4 PNK
3. Incubate in a thermocycler:
   - 37°C for 30 minutes (phosphorylation)
   - 95°C for 5 minutes (denaturation)
   - Ramp down to 25°C at 5°C/min (annealing)
4. Dilute annealed oligos 1:200 in nuclease-free water (final: 50 nM)

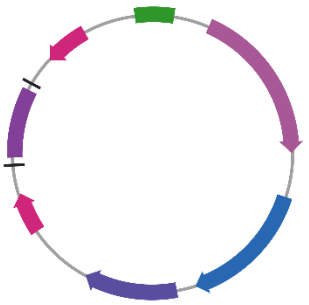
3. Golden Gate Assembly
1. Prepare reaction:
   - 100 ng lentiCRISPR v2 blast plasmid
   - 1 µL diluted annealed oligos (50 nM)
   - 2 µL 10X T4 DNA Ligase Buffer
   - 1 µL BsmBI-v2 (NEB)
   - 1 µL T4 DNA Ligase
   - Nuclease-free water to 20 µL total
2. Incubate in a thermocycler:
   - 37°C for 5 minutes
   - 16°C for 5 minutes
   - Repeat steps 1-2 for 10 cycles
   - 55°C for 5 minutes (final digestion)
   - 80°C for 5 minutes (enzyme inactivation)

4. Bacterial Transformation
1. Add 2 µL of the Golden Gate reaction to 50 µL of competent Stbl3 E. coli cells
2. Incubate on ice for 30 minutes
3. Heat shock at 42°C for 45 seconds
4. Return to ice for 2 minutes
5. Add 950 µL of SOC medium
6. Incubate at 37°C for 1 hour with shaking
7. Plate 100 µL on LB agar plates with ampicillin (100 µg/mL)
8. Incubate plates at 37°C overnight

5. Colony Screening and Verification
1. Pick 3-5 colonies and grow in 5 mL LB + ampicillin medium overnight
2. Extract plasmid DNA using a miniprep kit
3. Verify correct insertion by Sanger sequencing using the U6 promoter forward primer:
   5'-GACTATCATATGCTTACCGT-3'
4. Confirm the presence of the B2M sgRNA sequence in the plasmid

6. Lentivirus Production and Cell Transduction
1. Transfect the verified plasmid into HEK293T cells along with packaging plasmids
2. Collect virus-containing supernatant after 48-72 hours
3. Filter and concentrate the lentivirus if needed
4. Transduce target cells with the lentivirus
5. Select transduced cells with blasticidin (5-10 µg/mL)
6. Verify B2M knockout by flow cytometry, Western blot, or genomic sequencing

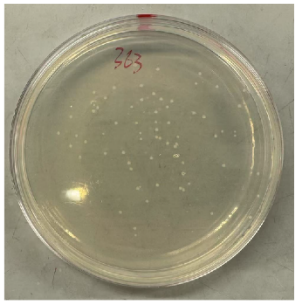
Plasmid Map

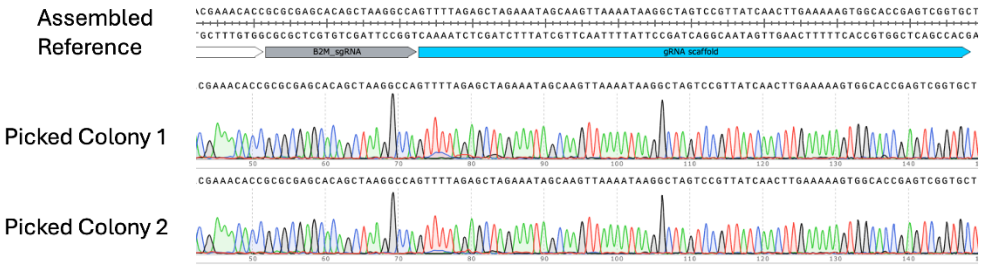




A scientist follows the protocol to perform the cloning

E





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Fig. 4. Biomni designs wet-lab experimental protocol. (**A**) Open-ended cloning benchmark on 10 real cloning scenarios. We compared against base LLM, trainee-level human, and expert-level human scientists. We found that Biomni has similar accuracy as the expert level scientist, and higher accuracy than trainee level, while using much less time. (**B**) Example of a user request to Biomni for cloning an sgRNA targeting the human B2M gene into the lentiCRISPR v2 Blast plasmid. (**C**) Biomni's automated stepwise workflow, including plasmid analysis, sgRNA design, oligo synthesis, Golden Gate assembly, bacterial transformation, colony screening, and final plasmid mapping. (**D**) Biomni-generated detailed cloning protocol with step-by-step instructions and comprehensive plasmid map, enabling laboratory scientists to execute the experiment autonomously. (**E**) Validation of Biomni's cloning protocol through successful colony growth on selection plates, followed by Sanger sequencing confirming perfect alignment of sgRNA insertion in picked colonies, demonstrating Biomni's robust capability for precise and reliable experimental design.

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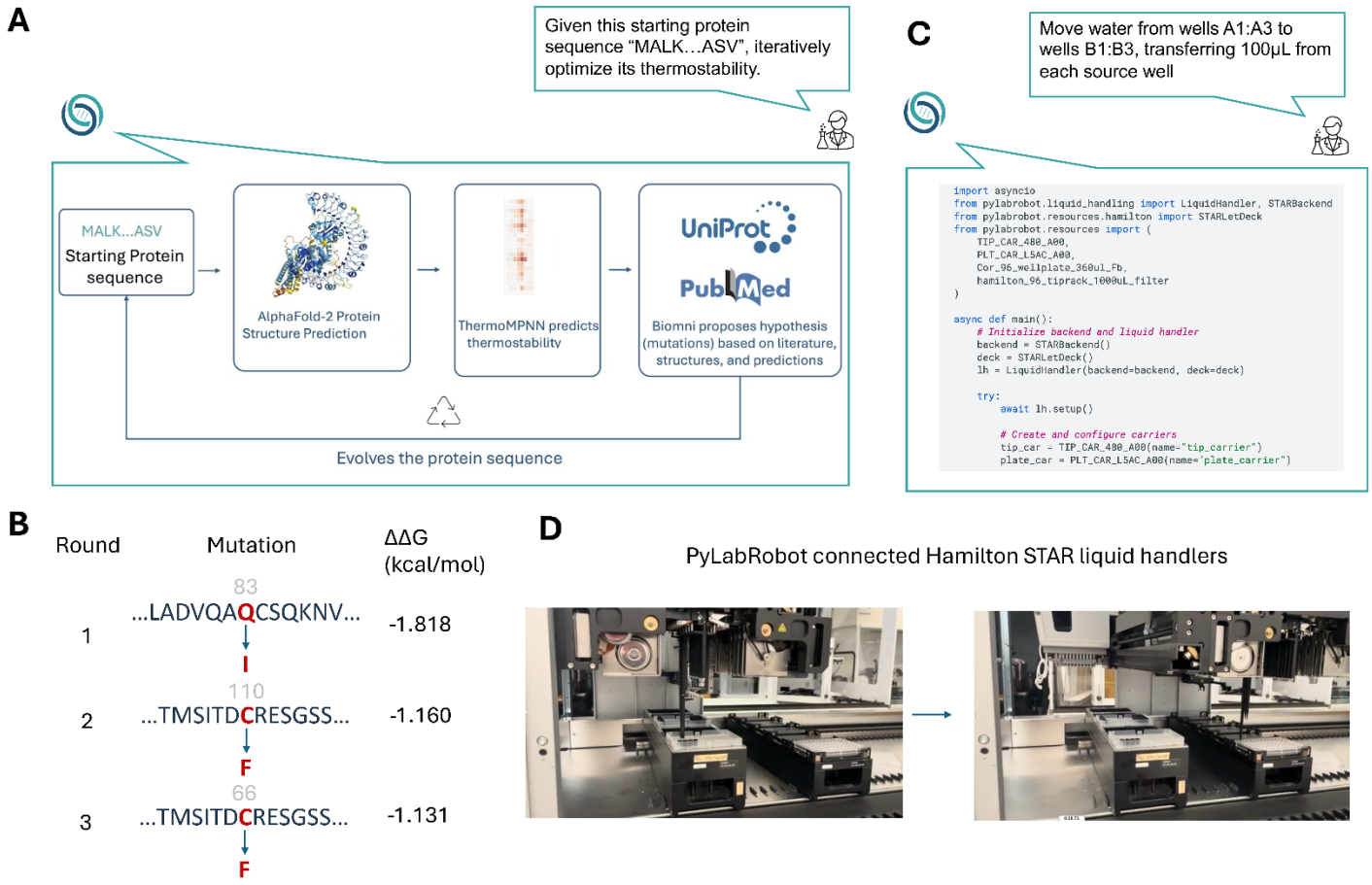


Fig. 5. AI-guided protein optimization and automated experimental execution. (**A**) Computational protein evolution pipeline. Biomni integrates structure prediction (AlphaFold-2), thermostability estimation (ThermoMPNN), and literature mining (UniProt/PubMed) to iteratively optimize a starting sequence ("MALK…ASV") for improved thermostability. (**B**) Example optimization trajectory. Over three iterative rounds, Biomni proposed stabilizing mutations (Q83I, C110F, C66F) yielding cumulative predicted ΔΔG improvement of -4.108 kcal/mol, corresponding to enhanced core hydrophobicity and aromatic stabilization while maintaining 98% sequence identity. (**C**) AI-driven automation of experimental protocols. Biomni converts natural language descriptions of lab tasks (e.g., "Move water from wells A1-A3 to B1-B3, transferring 100 µl each") into fully executable PyLabRobot code. (**D**) The code was converted and validated on Hamilton STAR platforms for end-to-end digital-to-physical experiment execution.

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Autonomous biomedical research with an artificial intelligence agent

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